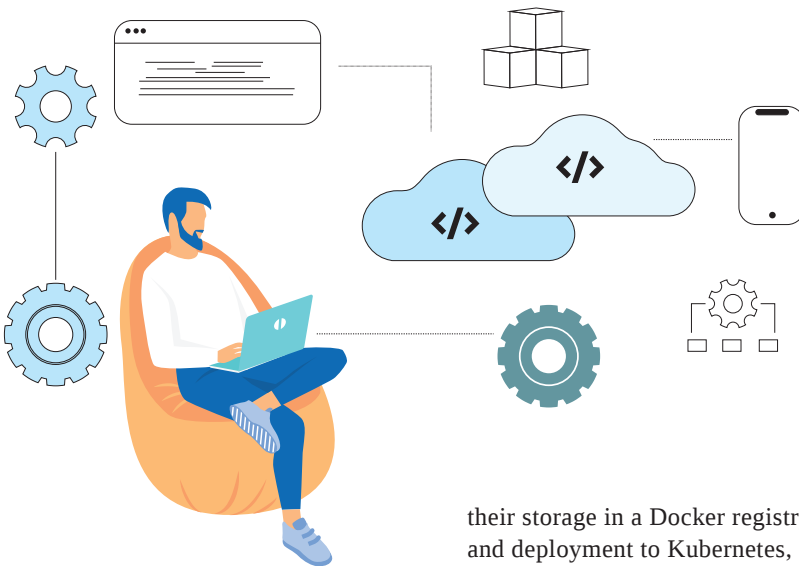


Deploying Jenkins X for Enhanced CI/CD and Cloud-Native Transition

Jenkins X is an open source project that extends the well-known Jenkins automation server to enable Kubernetes and cloud-native application development. It includes a suite of tools and best practices for automating CI/CD pipelines for cloud-native apps. Let's learn how to install and run it.



for managing most interactions with the platform. For seasoned Kubernetes users, 'kubectl' remains relevant for other tasks, while 'jx' takes charge of JX-specific activities, such as CI/CD and managing build logs. The process begins by using the 'jx' command line tool to create or import a project. JX then identifies the project type, generates a relevant Jenkinsfile if needed, specifies necessary Kubernetes resources (e.g., Helm charts), adds the project to GitHub, configures essential webhooks, initiates the application build in Jenkins, and deploys it to a staging environment upon successful tests. This integration establishes a fully operational CI/CD pipeline for the Kubernetes application.

Interactions with JX are primarily controlled through 'jx' commands for tasks like environment setup, application creation or import, and monitoring build pipeline status. Once configured, JX operates discreetly in the background, providing continuous CI/CD capabilities. This hands-off approach allows users to minimise direct interaction, as JX efficiently manages the intricacies of the CI/CD pipeline once the initial setup is complete.

The system relies on Kubernetes and Docker, where pods represent deployed microservices. Environments, managed through GitOps methodology, serve as virtual infrastructure stages

Developed by James Strachan, the creator of Groovy and Apache Camel, Jenkins X (JX) was introduced in March 2018. It streamlines the setup of Kubernetes environments, CI/CD pipelines, and GitHub integration for developers, essentially managing these tasks effortlessly.

Users may install all the tools they need for application administration, pipeline creation, and deployment across various environments with a single command. Notably, JX is pre-configured with Jenkins, avoiding the need for users to be familiar with certain plugins while setting up a CI/CD pipeline. JX automates the generation of Docker containers,

their storage in a Docker registry, and deployment to Kubernetes, bringing critical applications into the Kubernetes environment. The procedure is further expedited by JX's automation of Jenkins pipeline builds using Jenkinsfiles, which it may produce for current or new projects. In essence, Jenkins X enables customers to use CI/CD without prior knowledge of Kubernetes or Jenkins, since it manages the complexities of the full development and deployment life cycle.

Jenkins architecture

Building blocks

The primary interface with Kubernetes is the command line tool 'jx', offering a user-friendly alternative to 'kubectl'